

Unleashing the Potential of Vision-Language Models for Generalizable AI-Generated Image Detection

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Abstract

Recent work has shown that a simple linear probe on frozen representations from modern vision foundation models (VFMs) can achieve state-of-the-art AIGI detection performance, substantially outperforming specialized detectors in challenging in-the-wild scenarios. This finding has established DINOv3 as the dominant foundation-model baseline for subsequent improvements. However, we find that the vision-language model Perception Encoder (PE) holds greater potential for AIGI detection, because its language-aligned representation preserves high-level provenance semantics. Specifically, PE exhibits stronger local provenance organization than DINOv3 in its frozen feature space. However, semantic-agnostic linear probing fails to exploit this structure, as PE-Linear still underperforms DINOv3-Linear by 4.1% on In-the-Wild. Based on this observation, we propose Semantic Prototype Calibration (SPC), which constructs category prototypes from forensic semantic information and calibrates them with supervised data. We apply SPC to PE and refer to the resulting detector as PE-SPC. Our analysis shows that this simple design achieves stronger generalization. Across cross-generator, post-processing, and in-the-wild benchmarks, PE-SPC surpasses the previous DINOv3 baseline and achieves new state-of-the-art results.

Introduction

Synthetic images are becoming increasingly realistic and easy to produce, making reliable AIGI detection more important for maintaining trust in digital visual content. Conventional AIGI detectors achieve near-saturated accuracy on controlled benchmarks, yet their performance drops markedly in real-world scenarios. This is because limited training sources can lead models to rely on shortcuts, such as generator-specific textures or frequency anomalies, rather than learn generalizable differences between real and AI-generated images. New generators or common operations such as compression, blur, and resampling can alter these cues, causing detectors to fail.

On the other hand, Simplicity Prevails (Zhou et al. 2026) shows that modern VFMs, after exposure to abundant generated images during pretraining, can already distinguish real from generated content. Consequently, modern VFMs can outperform traditional detectors with only a linear probe.

Among the encoders evaluated in Simplicity Prevails, DINOv3 (Siméoni et al. 2025) achieves the strongest overall performance, outperforming vision-language models such as Perception Encoder (PE) (Bolya et al. 2025) on most benchmarks. Recent studies have likewise focused on purely visual models by further improving DINOv3-based detectors (Huang et al. 2025; Li et al. 2026).

However, we argue that this focus underestimates the potential of PE and, more broadly, of vision-language models for AIGI detection. As shown in Figure 1(a), DINOv3 is pre-trained on images alone, whereas PE learns from image-text pairs whose text may explicitly describe how an image was produced. Our comparison uses four In-the-wild Datasets, namely Chameleon (Yan et al. 2025), WildRF (Cavia et al. 2024), SocialRF (Li et al. 2025), and CommunityAI (Li et al. 2025). We randomly sample an equal number of features from each dataset and pool them, using 80% to fit a one-dimensional LDA projection (Fisher 1936) and reserving the remaining 20% for evaluation. We quantify class separation using the overlap coefficient (Inman and Bradley 1989), which measures the shared area between the real and generated LDA score distributions, with lower values indicating clearer separation. PE yields a substantially lower overlap coefficient than DINOv3, indicating a clearer forensic class structure. Following prior work (Zhou et al. 2026), we train both linear heads on GenImage SD1.4 (Zhu et al. 2023). PE-Linear nevertheless achieves lower average accuracy than DINOv3-Linear on the In-the-wild Datasets, showing that standard linear probing does not fully exploit this structure.

This gap arises because prior work uses only PE’s image encoder, discarding its text encoder and leaving text-derived forensic concepts unused. To incorporate these concepts into the classification head, we treat the two columns of its weight matrix as learnable forensic category prototypes for generated and real images, respectively. We introduce Semantic Prototype Calibration (SPC), which first uses PE’s text branch to provide a semantic starting point for these prototypes, and then calibrates them on the training data. This semantic starting point places the prototypes closer to task-optimal directions in PE’s feature space, shortening the optimization path required for calibration.

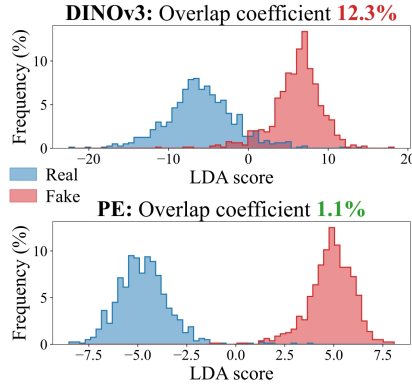
SPC consistently improves PE’s detection performance

(a) Provenance Cues in Pretraining



PE's image-text pairs may contain explicit textual provenance cues.

(b) Feature Distribution



PE features exhibit stronger provenance structure

(c) Linear-Probe Accuracy

	DINOv3-Linear	PE-Linear
Chameleon	91.4	95.9
ComminutyAI	94.8	97.1
SocialRF	94.3	86.1
WildRF	96.1	83.6
Avg.	94.0	89.9

Semantic-agnostic linear probing fails to exploit PE's provenance structure.

Yet PE-Linear underperforms DINOv3-Linear.

Figure 1: Our motivation is that standard linear probing fails to fully exploit the forensic structure already present in PE's features. (a) DINOv3 uses image-only pretraining, whereas PE learns from image-text pairs that may contain explicit provenance cues. (b) Using pooled features from the four In-the-wild Datasets, PE exhibits substantially less overlap between the real and generated LDA score distributions than DINOv3 (1.1% versus 12.3%). (c) When both linear heads are trained on GenImage SD1.4, PE-Linear nevertheless underperforms DINOv3-Linear on the In-the-wild Datasets (89.9% versus 94.0%).

across cross-generator, post-processing, and in-the-wild evaluations. PE-SPC achieves state-of-the-art performance on multiple benchmarks. These results highlight the promise of vision-language models for generalizable AIGI detection.

Our contributions are summarized as follows:

- We show that PE's performance under standard linear probing does not fully reflect its potential for AIGI detection. Its frozen features already exhibit clear forensic structure, yet the linear head fails to fully exploit it.
- We propose Semantic Prototype Calibration (SPC), which uses text-derived forensic semantics to guide the learning of category prototypes.
- We demonstrate the promise of vision-language models for AIGI detection. PE-SPC achieves state-of-the-art performance across multiple benchmarks with fewer parameters than DINOv3.

Related Work

Foundation-model-based AIGI detection. UnivFD (Ojha, Li, and Lee 2023) marked a shift toward foundation-model-based AIGI detection by showing that a linear classifier trained on a frozen CLIP image encoder can generalize across generators. AIDE (Yan et al. 2025), OMAT (Zhou et al. 2025a), DDA (Chen et al. 2025b), and Effort (Yan et al. 2024) further adapted pretrained VFMs to AIGI detection through task-specific architectures or training strategies.

More recently, Simplicity Prevails (Zhou et al. 2026) and VFM (Zhou et al. 2025b) showed that simple linear probes on modern VFMs can outperform specialized detectors, while SSAFE (Lee et al. 2026) and TAP (Abdullah, Ebert, and Wasenmüller 2026) further developed this frozen-feature paradigm. Among the encoders evaluated by Simplicity Prevails, DINOv3 (Siméoni et al. 2025) achieved the strongest

overall performance and outperformed vision-language models under the same linear-probing setting on most benchmarks. This strong baseline has motivated recent work to focus more on purely visual models, with FGTS (Huang et al. 2025) and STAL (Li et al. 2026) further improving DINOv3 through token selection and spectral auxiliary learning, respectively.

Vision-language models for AIGI detection. Vision-language models offer a complementary route because image-text pretraining can associate visual patterns with high-level concepts that describe how an image was produced. Existing CLIP-based detectors (Tan et al. 2025; Xu et al. 2025; Shi et al. 2025) demonstrate that language-aligned representations can improve generalization to unseen generators. However, these methods use task-specific training to inject textual concepts into or align them with the visual representation. In contrast, SPC requires no such representation adaptation and directly uses the forensic semantics already encoded in PE to guide the learning of category prototypes.

Geometric interpretation of linear classifiers. Conventional linear probing usually treats the classification head as a simple prediction layer on top of a VFM. However, a line of work in face recognition (Liu et al. 2017; Wang et al. 2017, 2018) offers a different geometric view. Each column of the classifier's weight matrix corresponds to one class, and its dot product with the image feature gives the score for that class. When the feature and class weights are normalized, this dot product becomes cosine similarity, allowing each column to be understood as a reference direction for its class. We adopt this view for AIGI detection and treat the two columns of a binary classifier's weight matrix as a pair of learnable forensic category prototypes. SPC uses the forensic semantics already encoded in PE to provide these prototypes

with task-relevant starting points before calibration.

Method

This section introduces SPC, focusing on how it uses text-derived forensic semantics to improve linear probing on a frozen PE encoder. We first interpret the two columns of a binary linear head as category prototypes, then examine the forensic information encoded in PE representations and why conventional linear probing fails to fully exploit it, and finally explain how SPC calibrates these prototypes.

Provenance Semantics in Vision-Language Representations

Prior work (Zhou et al. 2026, 2025b; Lee et al. 2026) has shown that modern visual encoders already contain the image-provenance information needed for AIGI detection. Unlike content semantics, provenance information describes *how an image was produced rather than what it depicts*, making it more difficult to learn from visual data alone. As shown in Figure 1(a), DINOv3 (Siméoni et al. 2025) is pretrained on images alone, whereas PE (Bolya et al. 2025) is pretrained on image-text pairs whose text may explicitly describe image provenance. In Figure 1(b), PE’s frozen features exhibit a clearer local forensic structure than DINOv3’s across diverse test sources. Appendix E reports the corresponding distributions for each In-the-Wild dataset separately.

Language alignment also enables a direct textual interpretation of the forensic information encoded in PE’s image features. Simplicity Prevails (Zhou et al. 2026) demonstrates this by comparing image features with a pool of text concepts, finding that generated images align with forgery-related concepts in PE’s shared embedding space. Because DINOv3 lacks a text encoder, it can not provide the same text-based interpretation of its forensic information.

Conventional Linear Probing Underestimates PE

Despite its clearer local forensic structure, PE still underperforms DINOv3 under conventional linear probing. As shown in Figure 1(c), when both linear heads are trained on Gen-Image SD1.4 (Zhu et al. 2023), PE-Linear achieves an average accuracy of 89.9 on In-the-Wild, compared with 94.0 for DINOv3-Linear. This discrepancy indicates that conventional linear probing fails to fully exploit the forensic structure already present in PE’s features.

Conventional linear-probing baselines (Ojha, Li, and Lee 2023; Zhou et al. 2026) typically freeze the image encoder, and train a randomly initialized linear head on a single source. When this setup is applied to a vision-language model, the text encoder remains unused. Consequently, the two category prototypes are learned solely from that source, and cannot benefit from the provenance semantics acquired through language alignment.

Therefore, better exploiting the AIGI detection potential of vision-language models does not require a more complex detector. The key is to change how the two forensic category prototypes are learned. Rather than estimating them entirely from a single training source, we use the model’s existing

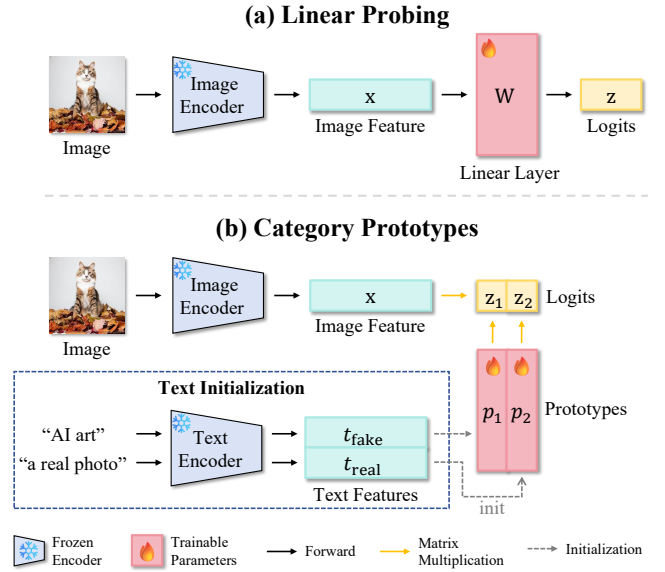


Figure 2: Category-prototype view of linear probing and Semantic Prototype Calibration (SPC). (a) Conventional linear probing uses a frozen image encoder to extract feature $\mathbf{x}$ and a trainable binary linear head W to produce logits $\mathbf{z}$. (b) Since $W = [p_1, p_2]$, its two columns can equivalently be viewed as trainable category prototypes, whose dot products with $\mathbf{x}$ produce logits z_1 and z_2 . SPC uses the text features of “AI art” and “a real photo” as semantic starting points for p_1 and p_2 . The text encoder is used only for initialization.

semantic information to provide a task-relevant starting point and then calibrate the prototypes on the training data.

Viewing the Classifier as Category Prototypes

Let f_I denote the frozen image encoder, $\mathcal{N}$ the ℓ_2 -normalization operator, and W the trainable binary linear head. Given an input image I , standard linear probing can be written as:

$$\mathbf{x} = \mathcal{N}(f_I(I)), \quad \mathbf{z} = \mathbf{x}W. \quad (1)$$

In a binary linear head, each column of the weight matrix W corresponds to one output class. We denote these columns by p_1 and p_2 , corresponding to generated and real images, respectively:

$$W = [p_1, p_2], \quad \mathbf{z} = [\mathbf{x}p_1, \mathbf{x}p_2]. \quad (2)$$

As illustrated in Figure 2, the binary linear head can therefore be viewed as a pair of learnable forensic category prototypes.

This prototype interpretation reveals a limitation of conventional linear probing. Without task-relevant starting points, p_1 and p_2 are learned solely from a single training source, and may therefore become biased toward patterns specific to that source. As a result, these prototypes may fail to capture the general forensic differences between generated and real images, limiting generalization to unseen generators and diverse real-image sources.

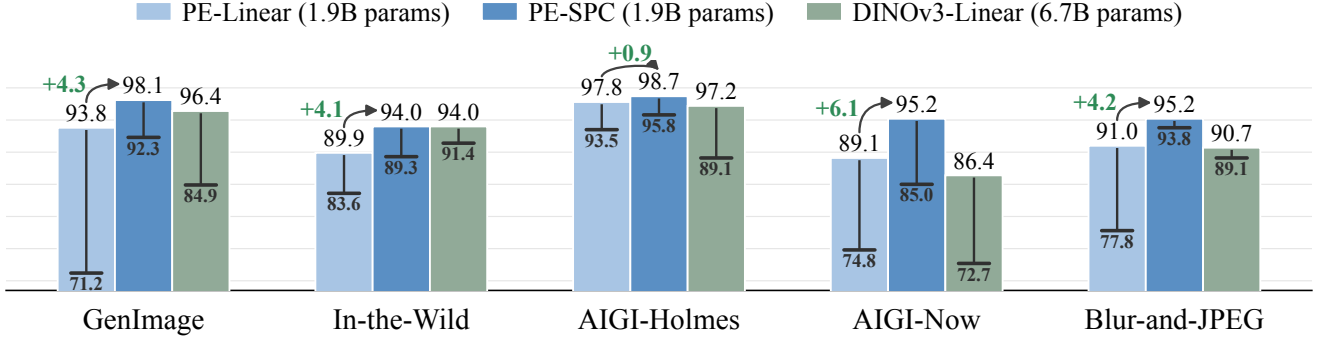


Figure 3: Average and minimum accuracy of PE-Linear, PE-SPC, and DINOv3-Linear across five benchmark groups. Bar heights show the average accuracy within each group, while black horizontal markers show the minimum accuracy over its constituent datasets. Model parameter counts are reported in the legend. SPC improves PE both metrics across all five groups.

Semantic Prototype Calibration

Unlike vision-only models, vision-language models align image and text representations in a shared embedding space (Radford et al. 2021). We therefore derive semantic starting points for p_1 and p_2 by encoding descriptions of generated and real images. Using the text encoder f_T , we compute and normalize the resulting text embeddings.

$$\begin{aligned} p_1 &\leftarrow \mathcal{N}(f_T(\text{"AI art"})), \\ p_2 &\leftarrow \mathcal{N}(f_T(\text{"a real photo"})). \end{aligned} \quad (3)$$

We then calibrate the two prototypes on the training set by minimizing the cross-entropy loss $\mathcal{L}_{CE}([xp_1, xp_2], y)$, while keeping the image encoder frozen.

We refer to this process as *prototype calibration* because training refines prototypes that already encode class semantics. The text-derived prototypes lie near task-optimal directions in PE’s embedding space, so calibration requires only limited adjustment. Such limited adjustment reduces the classifier’s reliance on training-source-specific cues, thereby improving cross-distribution generalization.

Once the semantic starting points are obtained, neither text input nor the text encoder is required for prototype calibration or inference. Consequently, PE-SPC retains the same inference architecture as PE-Linear, comprising a frozen image encoder and a binary linear head, and incurs no additional memory or computational cost at inference time.

Results

Experimental Setup

Benchmarks. We evaluate generalization across five benchmark groups covering unseen generators, in-the-wild data, recent generators, and common post-processing operations. GenImage (Zhu et al. 2023) evaluates cross-generator generalization using the seven generator subsets other than Stable Diffusion v1.4, which is used for training. In-the-Wild Datasets comprises Chameleon (Yan et al. 2025), WildRF (Cavia et al. 2024), and SocialRF and CommunityAI from AIGIBench (Li et al. 2025). These images were collected from social media or online communities, with un-

known generation sources and processing histories. AIGI-Holmes (Zhou et al. 2025c) and AIGI-Now (Chen et al. 2025a) evaluate generalization to recent generators. Blur-and-JPEG evaluates robustness to Gaussian blur and JPEG compression applied to Chameleon dataset.

Baselines. We evaluate SPC primarily on PE (Bolya et al. 2025), while MetaCLIP2 (Chuang et al. 2025), SigLIP2 (Tschannen et al. 2025), and OpenCLIP (Cherti et al. 2023) are included to assess its applicability across vision-language models. Because DINOv3 (Siméoni et al. 2025) achieved the strongest overall performance among the encoders evaluated in Simplicity Prevails (Zhou et al. 2026), we adopt it as a strong vision-only baseline. We additionally compare with DDA (Chen et al. 2025b), OMAT (Zhou et al. 2025a), and AIDE (Yan et al. 2025) as representative task-specific VFM-based detectors.

Implementation details. For every VFM, the encoder remains frozen, and only the binary linear head is trained on the Stable Diffusion v1.4 subset of GenImage (Zhu et al. 2023). For conventional linear probing, we follow Simplicity Prevails (Zhou et al. 2026) and optimize the head for two epochs using AdamW, with a batch size of 128 and a learning rate of 1×10^{-3} . Images are resized and center-cropped to each encoder’s native input resolution, without additional data augmentation. For SPC, we use a learning rate of 2×10^{-5} while keeping all other settings unchanged. To stabilize optimization during the early stage of prototype calibration, the generated- and real-class bias terms are initialized to 1 and 0, respectively. Appendices A, C, and D provide the hardware and software environment, learning-rate analysis, and checkpoint details, respectively.

Overall Performance

Figure 3 shows that SPC improves PE-Linear across all five benchmark groups. The largest gain occurs on AIGI-Now (+6.1%), while In-the-Wild and Blur-and-JPEG improve by 4.1% and 4.2%, respectively. These benchmarks differ substantially from the SDv1.4 training distribution, making cues tied to the training generator less reliable. The consistent gains under these shifts indicate that SPC enables the clas-

sification head to make better use of the transferable forensic information in PE’s representation. SPC also improves GenImage by 4.3%, demonstrating stronger cross-generator generalization within a controlled benchmark. Even on the near-saturated AIGI-Holmes benchmark, SPC delivers a further improvement of 0.9%, showing that it remains effective when the baseline performance is already high.

Across the detailed benchmark results, SPC’s gains are concentrated on the datasets and conditions where PE-Linear performs poorly. SPC improves overall performance primarily by substantially raising the minimum accuracy in every benchmark group, with the largest increases on GenImage (+21.1%), Blur-and-JPEG (+16.0%), and AIGI-Now (+10.2%). This results in more stable performance across diverse datasets. Overall, PE-SPC matches or exceeds DINOv3-Linear in average accuracy across all benchmarks.

PE-SPC achieves these results with a 1.9B-parameter ViT-G/14 backbone, compared with the 6.7B-parameter ViT-7B/16 backbone used by DINOv3-Linear. This favorable trade-off between accuracy and model scale highlights vision-language models as a promising direction for efficient AIGI detection, while leaving substantial room to explore larger backbones. Appendix B provides detailed model-cost and inference-efficiency measurements.

Performance on Standard Benchmarks

Table 1 evaluates cross-generator generalization under the standard GenImage (Zhu et al. 2023) protocol. PE-Linear already performs nearly perfectly on the Stable Diffusion and Wukong subsets, but transfers less effectively to ADM and Midjourney. SPC preserves PE’s strong performance on the former subsets while substantially improving the latter, indicating that SPC primarily improves transfer beyond the single training source. PE-SPC substantially improves upon PE-Linear and surpasses DINOv3-Linear, achieving the highest average accuracy.

Method	ADM	BigGAN	Midjourney	VQDM	GLIDE	SD-v1.4	SD-v1.5	Wukong	Avg.
DDA	88.0	74.1	96.0	71.9	86.2	98.6	98.5	98.6	89.0
OMAT	83.7	97.3	90.3	95.4	97.4	97.4	97.3	97.5	94.6
AIDE	60.2	64.8	81.3	69.4	51.4	95.9	95.7	95.4	76.8
SigLIP2-Linear	87.0	92.4	87.9	95.4	95.1	99.5	99.5	99.4	94.5
MetaCLIP2-Linear	69.0	81.6	95.9	81.9	88.7	99.3	99.1	98.0	89.2
PE-Linear	71.2	96.3	90.1	95.9	97.2	99.9	99.9	99.9	93.8
DINOv3-Linear	84.9	99.1	93.4	99.2	96.3	99.8	99.6	99.4	96.4
PE-SPC (Ours)	92.3	99.3	95.8	99.4	98.4	99.8	99.7	99.7	98.1

Table 1: Accuracy on GenImage (Zhu et al. 2023). Bold denotes the best result only when it exceeds the second-best result by more than 0.2%. PE-SPC achieves the highest average accuracy of 98.1%, outperforming PE-Linear by 4.3% and DINOv3-Linear by 1.7%. Its largest gains over PE-Linear occur on ADM (+21.1%) and Midjourney (+5.7%), and it ranks first on four of the seven unseen-generator subsets.

Generalization to In-the-Wild Data

Table 2 evaluates robustness across four in-the-wild collections with heterogeneous sources, unknown generators, and uncontrolled processing histories. PE-Linear performs strongly on Chameleon and CommunityAI but transfers less effectively to SocialRF and WildRF, revealing substantial variation across real-world sources. SPC delivers its largest gains on the two weaker collections while retaining strong performance on Chameleon and CommunityAI, resulting in more balanced performance and matching DINOv3-Linear on average. This improved stability makes PE-SPC more reliable in practical settings, where image provenance and processing history are usually unknown.

Method	Cham.	CommAI	SocialRF	WildRF	Avg.
DDA	82.4	84.7	83.2	90.4	85.0
OMAT	62.9	63.4	60.7	67.4	63.6
AIDE	57.4	54.2	56.0	58.4	56.5
SigLIP2-Linear	85.9	86.6	80.5	79.0	82.2
MetaCLIP2-Linear	93.0	94.0	80.0	72.8	84.2
PE-Linear	95.9	97.1	86.1	83.6	89.9
DINOv3-Linear	91.4	94.8	94.3	96.1	94.0
PE-SPC (Ours)	94.7	97.2	89.3	94.8	94.0

Table 2: Accuracy across In-the-Wild, comprising Chameleon (Yan et al. 2025), WildRF (Cavia et al. 2024), and SocialRF and CommunityAI from AIGIBench (Li et al. 2025). “Cham.” and “CommAI” abbreviate Chameleon and CommunityAI, respectively. PE-SPC improves the overall accuracy of PE from 89.9% to 94.0%, matching DINOv3-Linear. Most of the gain comes from SocialRF (+3.2%) and WildRF (+11.2%).

Generalization to State-of-the-Art Generators

To test whether PE-SPC exploits transferable forensic information rather than generator-specific patterns memorized during image-encoder pretraining, we evaluate it on AIGI-Now (Chen et al. 2025a) and AIGI-Holmes (Zhou et al. 2025c), both of which include generators unseen during pre-training. AIGI-Now includes closed-source generators unseen during image-encoder pretraining, such as GPT-4o and FLUX-Pro, and separates evaluation into a “pix” split that isolates low-level generative traces by aligning image formats and a “sem” split that suppresses these traces through strong degradations to emphasize high-level semantic anomalies. In Tables 3, PE-SPC achieves the best overall performance on AIGI-Now, clearly surpassing DINOv3-Linear. Relative to PE-Linear, its improvement is substantially larger on “sem” than on “pix”, indicating that SPC primarily strengthens the use of high-level forensic semantics.

AIGI-Holmes extends this evaluation to recent generators whose architectures differ from the UNet-based SD1.4 training source, including autoregressive models such as LlamaGen and VAR, as well as diffusion transformers such as FLUX and SD3.5-L. On the near-saturated AIGI-Holmes benchmark in table 4, PE-SPC again ranks first and outperforms both PE-Linear and DINOv3-Linear. Compared

Detector	FLUX-dev		FLUX-kera		Kontext		FLUX-pro		gpt4o		jimeng		keling		minimax		Nano		Avg.
	pix	sem	pix	sem	pix	sem	pix	sem	pix	sem	pix	sem	pix	sem	pix	sem	pix	sem	
DDA	91.6	51.2	59.4	49.9	82.7	52.9	76.6	55.0	92.3	65.4	87.0	65.4	96.1	64.6	83.3	50.5	81.6	56.2	69.5
OMAT	91.1	47.5	64.9	46.9	84.7	50.7	59.1	51.5	74.4	45.2	49.1	46.5	93.6	52.6	69.9	46.7	89.1	46.8	61.5
AIDE	99.1	59.0	50.4	56.9	97.9	80.6	60.1	53.8	74.7	51.8	63.9	51.4	98.2	55.4	51.4	54.1	98.9	51.8	67.2
SigLIP2-Linear	94.7	88.2	88.3	69.7	77.6	67.8	88.8	88.5	93.6	79.0	83.1	84.5	94.1	86.7	85.0	68.8	89.5	88.2	84.3
MetaCLIP2-Linear	97.9	94.1	96.3	89.6	79.9	81.1	97.6	89.2	94.3	88.8	96.5	82.5	97.0	90.2	94.2	85.0	96.5	81.9	90.7
PE-Linear	97.7	95.9	91.8	76.2	83.0	77.4	87.3	94.3	86.3	92.4	91.5	92.1	93.9	91.6	86.5	74.8	97.1	93.6	89.1
DINOv3-Linear	94.4	96.2	84.6	81.1	73.0	75.6	81.3	94.8	89.8	96.0	82.4	94.0	88.4	91.3	72.7	78.4	89.8	92.2	86.4
PE-SPC (Ours)	99.0	98.0	96.3	95.8	90.5	91.6	86.7	97.0	98.0	99.2	85.0	98.7	97.0	96.7	94.9	94.5	97.6	97.8	95.2

Table 3: Accuracy on AIGI-Now (Chen et al. 2025a), where “Kontext” abbreviates FLUX-kontext. PE-SPC achieves the best overall accuracy of 95.2%, exceeding DINOv3-Linear by 8.8%. Its average gain over PE-Linear is substantially larger on sem than on pix (+9.0% vs. +3.3%), with representative improvements on GPT-4o (+11.7% pix and +6.8% sem), FLUX-Kontext (+7.5% pix and +14.2% sem), and MiniMax (+8.4% pix and +19.7% sem).

Detector	FLUX	Infinity	JP-1B	JP-7B	Janus	LGen	PA-XL	SD3.5-L	Show-o	VAR	Avg.
DDA	97.2	98.9	99.3	98.7	98.5	99.3	99.4	97.0	94.8	80.4	96.3
OMAT	94.7	95.5	75.6	64.1	65.1	96.7	96.9	95.7	96.9	95.9	87.7
AIDE	94.4	98.7	98.9	97.8	91.2	99.4	98.6	99.4	98.0	93.6	97.0
SigLIP2-Linear	95.7	99.4	99.3	98.9	99.0	99.1	99.4	91.4	99.2	91.3	97.3
MetaCLIP2-Linear	98.7	99.0	95.9	92.8	83.9	98.9	98.6	95.6	98.5	80.2	94.2
PE-Linear	96.8	99.9	99.5	99.6	94.5	100.0	100.0	94.3	99.9	93.5	97.8
DINOv3-Linear	93.3	99.8	99.5	98.6	99.6	99.9	99.9	89.1	99.7	92.2	97.2
PE-SPC (Ours)	98.8	99.8	99.1	97.6	95.8	99.6	99.7	98.7	99.6	98.3	98.7

Table 4: Accuracy on AIGI-Holmes (Zhou et al. 2025c), where “JP-1B,” “JP-7B,” “LGen,” and “PA-XL” abbreviate Janus-Pro-1B, Janus-Pro-7B, LlamaGen, and PixArt-XL, respectively. Bold denotes the best result only when it exceeds the second-best result by more than 0.2%. PE-SPC achieves the best overall accuracy of 98.7%, improving over DINOv3-Linear by 1.5%. Its gains over PE-Linear on VAR (+4.8%), SD3.5-L (+4.4%), and FLUX (+2.0%) demonstrate transfer across both autoregressive and diffusion-transformer generators.

with PE-Linear, its gains are concentrated on VAR, SD3.5-L, and FLUX, spanning both autoregressive and diffusion-transformer generators. Together with the AIGI-Now results, this transfer to generators unseen during pretraining and to architectures distinct from SD1.4 provides evidence that SPC exploits generalizable forensic information rather than memorized generator-specific patterns.

Resilience to Common Perturbations

Table 5 evaluates robustness to JPEG compression and Gaussian blur, two common post-processing operations encountered during image dissemination. Both perturbations weaken low-level forensic cues, as JPEG compression alters local image statistics through quantization and Gaussian blur progressively removes high-frequency information. PE-SPC achieves the highest average performance and outperforms DINOv3-Linear across all perturbation settings. PE-Linear remains strong under JPEG compression but deteriorates rapidly as the blur strength increases, whereas PE-SPC remains stable even under severe blur. This contrast indicates that SPC reduces reliance on post-processing-sensitive cues and makes better use of forensic information preserved under common perturbations.

Detector	JPEG Compression				Gaussian Blur				Avg.
	95	85	75	65	0.5	1.0	1.5	2.0	
DDA	83.6	82.3	79.0	79.0	84.4	84.3	79.7	75.8	81.0
SigLIP2-Linear	85.7	85.6	86.0	82.8	83.6	77.6	68.9	67.1	79.7
MetaCLIP2-Linear	93.3	94.0	93.7	89.8	92.2	93.1	93.9	93.2	92.9
PE-Linear	96.7	96.4	95.9	92.1	93.8	92.5	83.1	77.8	91.0
DINOv3-Linear	91.8	91.4	91.6	89.1	91.6	90.9	89.7	89.1	90.7
PE-SPC (Ours)	95.2	95.7	94.8	94.0	93.8	95.2	96.7	95.9	95.2

Table 5: Accuracy under JPEG compression and Gaussian blur on Chameleon images. JPEG uses quality factors 95, 85, 75, and 65, while Gaussian blur uses $\sigma = 0.5, 1.0, 1.5$, and 2.0. PE-SPC achieves the best average accuracy of 95.2%, improving over DINOv3-Linear by 4.5%. Under the strongest blur of $\sigma = 2.0$, it achieves 95.9% accuracy, exceeding PE-Linear by 18.1% and DINOv3-Linear by 6.8%.

Analysis

How Do Prompt Semantics Drive SPC’s Gains?

Table 6 examines whether SPC’s gains arise from exact word-ing, text-derived starting points alone, or forensic semantics

Setting	Fake-class	Real-class	GenImage	In-the-Wild	AIGI-Holmes	AIGI-Now	Blur-and-JPEG	Avg. Δ
No SPC	–	–	93.8	89.9	97.8	89.1	91.0	0.0
Our Prompt	AI art	a real photo	98.1	94.0	98.7	95.2	95.2	+3.9
Similar Prompt	AI-generated	real	98.5	90.3	98.9	97.6	94.8	+3.7
Opposite Prompt	a real photo	AI art	81.5	44.7	78.2	65.5	60.5	-26.2
Unrelated Prompt I	sunset	landscape	94.8	77.5	92.8	79.3	78.3	-7.8
Unrelated Prompt II	portrait	technology	93.8	75.6	92.3	70.2	84.1	-9.1

Table 6: Benchmark-level average accuracy of PE under different prompt settings for SPC. Avg. Δ denotes the mean change relative to No SPC across the five benchmark groups. Our Prompt uses the provenance descriptions adopted in our main experiments, whereas Similar Prompt uses semantically equivalent wording. Both improve all five groups, with average gains of 3.9% and 3.7%, respectively. Opposite Prompt swaps the class assignments of Our Prompt and reduces average performance by 26.2%. Unrelated Prompt I and Unrelated Prompt II use content descriptions unrelated to provenance and reduce average performance by 7.8% and 9.1%, respectively.

correctly aligned with the class labels. Our Prompt and Similar Prompt use semantically equivalent provenance descriptions and achieve comparable gains across all five benchmark groups, showing that SPC does not depend on exact wording. Opposite Prompt reverses the class assignments and sharply degrades performance, demonstrating that the forensic semantics must be aligned with the correct classes. Unrelated Prompt I and Unrelated Prompt II replace forensic descriptions with unrelated content semantics and both perform worse than No SPC, ruling out a generic benefit from text-derived starting points. Together, these results show that SPC works through class-aligned forensic semantics rather than prompt wording or text-derived starting points alone.

When Does a Vision-Language Model Benefit from SPC?

This section analyzes how SPC improves AIGI detection in vision-language models and why it benefits some models but not others. Figure 4 shows that SPC’s effect depends on whether a model has learned forensic information that distinguishes generated from real images during pretraining. Models that recognize synthetic images as AI-generated improve under SPC, whereas those that associate them with content concepts or incorrectly regard them as genuine deteriorate. This contrast demonstrates that SPC’s gains come not from text embeddings alone but from forensic information learned during pretraining. Appendix F details the candidate concepts and aggregation protocol used in this analysis.

The OpenCLIP checkpoint used in our experiments was pretrained on LAION-2B, which was collected in 2021–2022 before the rapid growth of synthetic content on the web (Schuhmann et al. 2022). Despite its recent release, SigLIP2 exhibits the same negative response to SPC as OpenCLIP, likely because it still relies on WebLI, which also predates this growth (Tschannen et al. 2025; Zhou et al. 2026). By contrast, MetaCLIP2 and PE were pretrained on newer web corpora containing large volumes of AI-generated images together with captions or metadata that reveal their origin (Chuang et al. 2025; Bolya et al. 2025).

The usefulness of SPC’s semantic starting points depends on whether the model learned relevant forensic information during pretraining. When synthetic images are paired with captions or metadata that identify them as AI-generated, the

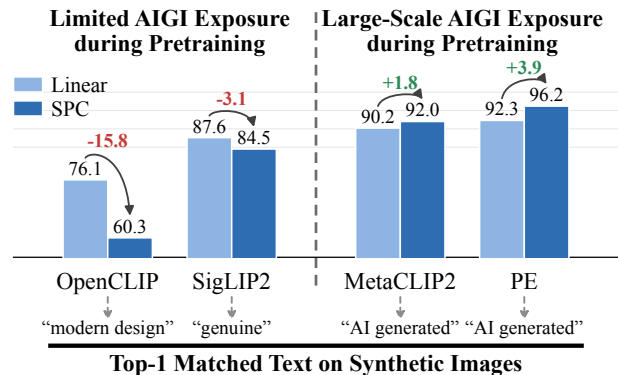


Figure 4: Effect of SPC on vision-language models with different levels of AIGI exposure during pretraining. The text below each model is the Top-1 match for synthetic Chameleon images from a pool of forensic-, content-, and source-related concepts. With limited AIGI exposure, OpenCLIP matches the concept “modern design” and SigLIP2 incorrectly matches the concept “genuine”, while both degrade under SPC. With large-scale AIGI exposure, MetaCLIP2 and PE both match “AI generated” and improve under SPC.

model learns to associate visual evidence of generation with forensic concepts. The resulting text-derived prototypes already lie near task-optimal directions, shortening the optimization path required for calibration. Without this association, the text-derived prototypes are misaligned with the model’s image representations and can instead steer calibration in the wrong direction.

Conclusion

We show that PE’s language-aligned representation contains clear forensic structure that conventional linear probing fails to exploit. SPC addresses this limitation by providing category prototypes with text-derived forensic starting points before calibration. Our analysis shows that its gains depend on both forensic information learned during pretraining and correct semantic alignment with the classes, rather than on text embeddings alone. Across cross-generator, post-processing,

and in-the-wild benchmarks, PE-SPC achieves state-of-the-art performance with fewer parameters than DINOv3, highlighting vision-language models as a promising direction for generalizable AIGI detection.

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Appendix

A Hardware and Software Environment

We implemented all experiments in PyTorch 2.2.0 and ran them on an NVIDIA Tesla V100S GPU with 32 GB of memory and an Intel Xeon Gold 6226R CPU. The software environment consisted of NVIDIA driver 510.47.03, CUDA 11.8, and cuDNN 8.7.

B Model Cost and Inference Efficiency

Throughout this appendix, the five benchmark groups refer to GenImage, In-the-Wild, AIGI-Holmes, AIGI-Now, and Blur-and-JPEG. As shown in Table 7, PE-SPC uses a substantially smaller backbone than DINOv3-Linear, yet achieves higher average accuracy, consumes less inference memory, and delivers higher throughput. These efficiency gains make PE-SPC better suited to resource-constrained environments and large-scale image screening. At inference, PE-SPC uses the same architecture as PE-Linear and does not require the text encoder, so SPC introduces no additional deployment overhead. Overall, these results highlight the potential of vision-language models for accurate and efficient AIGI detection.

Metric	DINOv3-Linear	PE-SPC	Δ
ViT Backbone	ViT-7B/16	ViT-G/14	–
Parameters	6.7 B	1.9 B	-71.6%
Memory Usage	26.1 GiB	12.5 GiB	-52.1%
Throughput	4.5 images/s	12.3 images/s	+173.3%
Accuracy	92.9	96.2	+3.6%

Table 7: Comparison of model size, inference efficiency, and average accuracy across the five benchmark groups. Memory usage and throughput are measured at a batch size of 32.

C Learning-Rate Sensitivity and Selection

Following Simplicity Prevails (Zhou et al. 2026), we use a learning rate of 1×10^{-3} for conventional linear probing. Unlike the randomly initialized head in conventional linear probing, SPC starts from text-derived prototypes that are already close to the target class directions in PE’s embedding space, so only small updates are needed for calibration. We therefore evaluate SPC across a range of learning rates, keeping the training data, optimizer, and batch size unchanged and the training duration fixed at two epochs.

Overall, 2×10^{-5} provides the best balance across the five benchmark groups. A smaller learning rate consistently reduces accuracy, whereas larger learning rates yield modest gains on several groups at the cost of a pronounced drop on In-the-Wild. We therefore set the learning rate to 2×10^{-5} in all main experiments.

D Pretrained Model Details

For each model family, we selected the largest publicly available checkpoint, reducing potential bias from the use of a smaller model variant. As shown in Table 8, OpenCLIP and

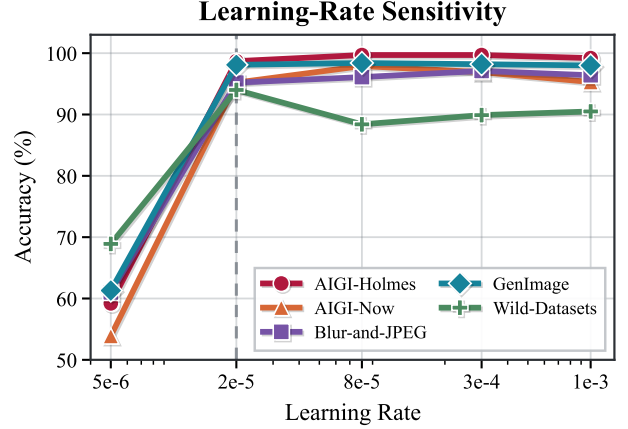


Figure 5: Learning-rate sensitivity of PE-SPC across the five benchmark groups after two epochs of prototype calibration. Each point reports the average accuracy within one group. Compared with the 1×10^{-3} setting used for conventional linear probing, 2×10^{-5} achieves comparable performance on four groups while improving In-the-Wild accuracy.

SigLIP2 rely on LAION and WebLI corpora released in 2022 and 2023, respectively (Cherti et al. 2023; Schuhmann et al. 2022; Tschannen et al. 2025), whereas DINOv3, PE and MetaCLIP2 use newer web corpora reported in their 2025 model papers (Siméoni et al. 2025; Bolya et al. 2025; Chuang et al. 2025). LAION and WebLI mostly contain web data collected before AI-generated images became widespread online. In contrast, the newer data used by DINOv3, PE and MetaCLIP2 contain many AI-generated images, together with captions or metadata that identify them as AI-generated. Consistent with this difference, SPC improves the overall performance of PE and MetaCLIP2 but reduces that of OpenCLIP and SigLIP2. Together, these results support the analysis in Figure 4, showing that SPC’s effect depends on whether the model learned the forensic information needed to distinguish real from generated images during pretraining.

E Per-Dataset Feature-Space Analysis

Figure 6 complements the pooled analysis in Figure 1(b) by analyzing the four In-the-Wild datasets Chameleon (Yan et al. 2025), WildRF (Cavia et al. 2024), and SocialRF and CommunityAI (Li et al. 2025) separately.

Across all four datasets, PE yields substantially lower overlap coefficients than DINOv3, with the relative difference approaching or exceeding an order of magnitude. This consistent pattern confirms the conclusion from Figure 1(b): PE’s frozen features exhibit clearer local forensic structure than DINOv3’s across diverse test sources.

Model	ViT Backbone	Params.	Input Res.	Output Dim.	Pretraining Data*
OpenCLIP	ViT-bigG/14	1.84B	224 × 224	1280	LAION-2B English (2022); LAION-Aesthetics (2022)
SigLIP2	ViT-g-opt/16	1.16B	384 × 384	1536	WebLI (2023)
MetaCLIP2	ViT-bigG/14	1.84B	378 × 378	1280	Worldwide public-web image-text corpus (2025)
Perception Encoder	ViT-G/14	1.88B	448 × 448	1280	Public-web image-text corpus and Video corpus (2025)
DINOv3	ViT-7B/16	6.716B	224 × 224	4096	LVD-1689M (2025); ImageNet-22K (2010); Mapillary (2020)

Table 8: Architecture, scale, and pretraining data of the foundation models evaluated in this work. “Params.” reports the parameter count of the visual backbone, “Input Res.” the evaluation resolution, and “Output Dim.” the dimensionality of the image features passed to the binary linear head. *Only the principal pretraining sources reported in the original model papers are listed, and the years in parentheses indicate publication years rather than exact data-collection cutoffs.

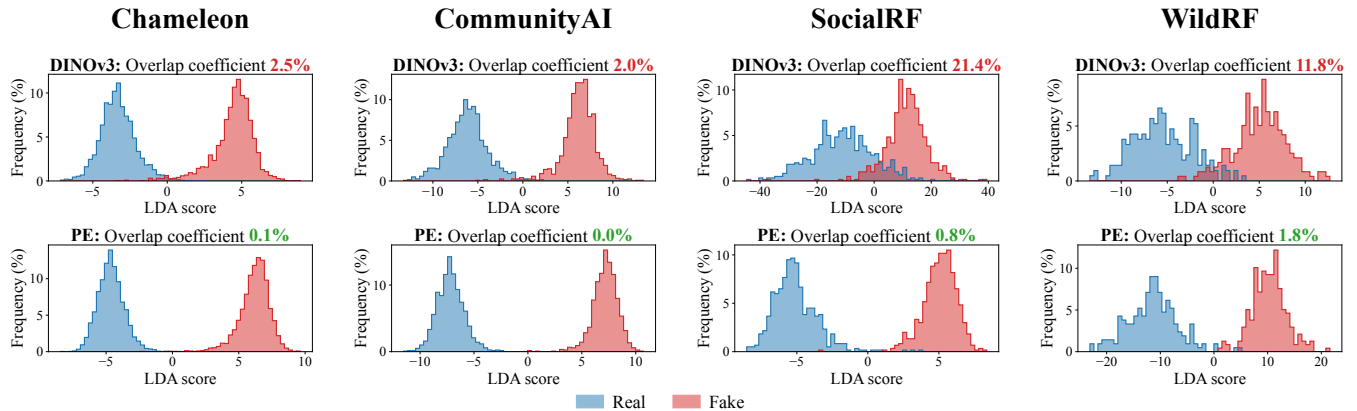


Figure 6: Per-dataset feature-space analysis of DINOv3 and PE, complementing the pooled analysis in Figure 1(b). Each column corresponds to one of the four In-the-Wild datasets and shows the one-dimensional LDA score distributions of real and generated images, obtained using the same fitting and held-out evaluation protocol as in Figure 1(b). The top and bottom rows show DINOv3 and PE, respectively. The overlap coefficient (Inman and Bradley 1989) measures the shared area between the two distributions, with lower values indicating clearer separation.

F Text-Concept Matching Protocol

Following the text-image matching protocol in Simplicity Prevails (Zhou et al. 2026), we organize the candidate text concepts into three groups:

- **Forensic concepts:** terms describing whether an image is real, manipulated, or AI-generated, including “fake,” “real,” “AI generated,” “authentic,” “genuine,” “manipulated,” and “synthetic.”
- **Content concepts:** terms describing what an image depicts without referring to how it was produced, including “sunset,” “landscape,” “portrait,” “abstract art,” “modern design,” “technology,” and “nature.”
- **Source concepts:** names of generative models, datasets, or platforms, including “GenImage,” “ADM,” “BigGAN,” “glide,” and “Midjourney.”

For each candidate concept, we compute the cosine similarity between its text embedding and the embedding of every synthetic image in Chameleon, then average the similarities across images. The concept with the highest average similarity is reported as the model-level Top-1 match in Figure 4. This protocol reveals whether a model recognizes synthetic images as AI-generated, associates them with unrelated content, or incorrectly regards them as genuine.